Introduction to Right Triangle Trigonometry

Introduction to Right Triangle Ratios

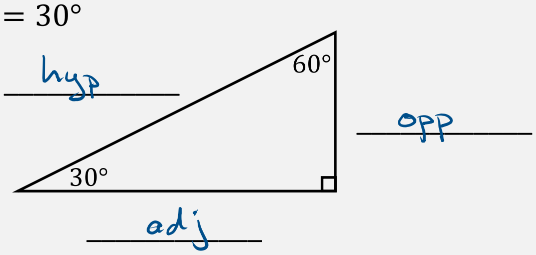
In a recent lesson, we reviewed similar triangles. We ended that discussion by investigating the ratio between any two sides of a triangle. Here, we'll focus our attention on these ratios, specifically in the case of a right triangle. Informally, we discussed labeling the sides of a triangle as small, medium, and large. Moving forward, we must develop a more formal technique to identify these sides.

To begin with, we first note that in any right triangle, there will always be two acute angles. (*Why?*) We select one of these angles as a reference. Then, we can identify the side opposite this angle as the __opposite__ side. The side across from the right angle is always called the __hypotenuse__. This leaves the remaining side, which is next to the referenced angle as our __adjacent__ side.

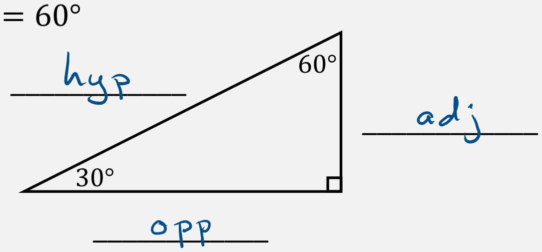
IMPORTANT INFO: ***Identifying Sides of a Right Triangle***

Using the triangles below, identify the *hypotenuse*, *opposite*, and *adjacent* sides to the referenced angle, θ.

Referenced angle
$\theta = 30°$

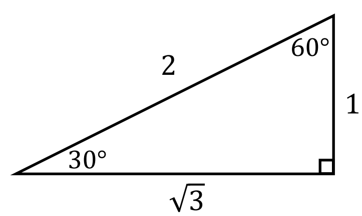


Referenced angle
$\theta = 60°$



EXAMPLE 1

Using the angle θ as a reference, evaluate each requested ratio using the right triangle shown here.



$\theta = 30°$, $\frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{2}$

$\theta = 60°$, $\frac{\text{opposite}}{\text{hypotenuse}} = \frac{\sqrt{3}}{2}$

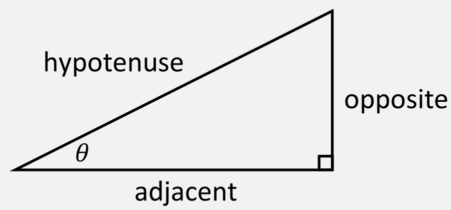
$\theta = 30°$, $\frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\sqrt{3}}{2}$

$\theta = 60°$, $\frac{\text{adjacent}}{\text{hypotenuse}} = \frac{1}{2}$

$\theta = 30°$, $\frac{\text{opposite}}{\text{adjacent}} = \frac{1}{\sqrt{3}}\left(\frac{\sqrt{3}}{\sqrt{3}}\right) = \frac{\sqrt{3}}{3}$

$\theta = 60°$, $\frac{\text{opposite}}{\text{adjacent}} = \frac{\sqrt{3}}{1} = \sqrt{3}$

In the last example, we see it takes a good deal of writing to communicate precisely which ratio we were interested in finding. We now introduce three trigonometric functions that provide a short-hand method to identify any ratio for the side measures in a right triangle.

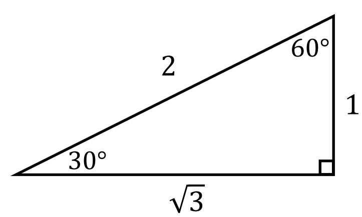
DEFINITION: ***Right Triangle Based Definitions for Sine, Cosine, and Tangent***	
Sine: $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$	
Cosine: $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$	
Tangent: $\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$	
Note: There are three other ratios possible. In particular, the reciprocal ratios to those provided here. We'll save these for another lesson.	

We can use the acronym, SOH - CAH - TOA, to help us recall the above ratios.

SOH – CAH – TOA		
$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$	$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$	$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$

EXAMPLE 2

Using the right triangle trigonometric definitions above, evaluate each of the following.



SOH - CAH - TOA

SOH $\sin(30°) = \frac{\text{opp}}{\text{hyp}} = \frac{1}{2}$

$\sin(60°) = \frac{\sqrt{3}}{2}$

CAH $\cos(30°) = \frac{\text{adj}}{\text{hyp}} = \frac{\sqrt{3}}{2}$

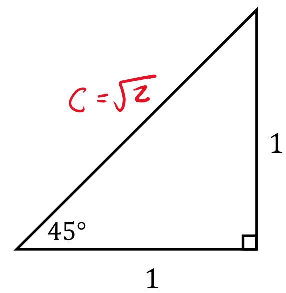
$\cos(60°) = \frac{1}{2}$

TOA $\tan(30°) = \frac{\text{opp}}{\text{adj}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

$\tan(60°) = \frac{\sqrt{3}}{1} = \sqrt{3}$

EXAMPLE 3

Using the right triangle trigonometric definitions, find the sine, cosine, and tangent values indicated below.



SOH $\sin(45°) = \frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{\sqrt{2}}{2}$

CAH $\cos(45°) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

TOA $\tan(45°) = \frac{1}{1} = 1$

$c^2 = 1^2 + 1^2$
$c^2 = 2$
$c = \sqrt{2}$

Trigonometric Function Values for Special Angles

The last two examples demonstrate some of the most common values you'll be working with in this course. These results, and a few others, have been summarized in the following table. We'll discuss a quick way of reproducing this table whenever needed, *without memorization*!

Trigonometric Function Values for Special Angles					
	$\boldsymbol{\theta = 0°}$	$\boldsymbol{\theta = 30°}$	$\boldsymbol{\theta = 45°}$	$\boldsymbol{\theta = 60°}$	$\boldsymbol{\theta = 90°}$
$\mathbf{sin}\,\boldsymbol{\theta}$	0	$^1/_2$	$^{\sqrt{2}}/_2$	$^{\sqrt{3}}/_2$	1
$\mathbf{cos}\,\boldsymbol{\theta}$	1	$^{\sqrt{3}}/_2$	$^{\sqrt{2}}/_2$	$^1/_2$	0
$\mathbf{tan}\,\boldsymbol{\theta}$	0	$^{\sqrt{3}}/_3$	1	$\sqrt{3}$	undefined

EXAMPLE 4

Practice reproducing the above table by completing the following steps.

1. All entries for sine are of the form, $\sqrt{n}/2$. Start with $n = 0$ and count up.
2. For cosine, reverse the sine values.
3. For tangent, we'll use our first identity, $\tan\theta = \sin\theta\,/\,\cos\theta$ (more on this later).

	0°	30°	45°	60°	90°
$\sin\theta$	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$

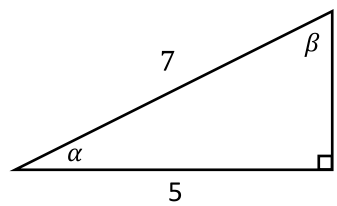
	0°	30°	45°	60°	90°
$\sin\theta$	0	$^1/_2$	$^{\sqrt{2}}/_2$	$^{\sqrt{3}}/_2$	1
$\cos\theta$	1	$^{\sqrt{3}}/_2$	$^{\sqrt{2}}/_2$	$^1/_2$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.

reverse order

**Reproducing this table is an important skill you'll use throughout this text.*

EXAMPLE 5

Using the provided triangle, evaluate each of the following.



$b = 2\sqrt{6}$

$5^2 + b^2 = 7^2$

$b^2 = 24$

$b = \sqrt{24}$

$b = \sqrt{4 \cdot 6} = 2\sqrt{6}$

SOH $\sin(\alpha) = \frac{2\sqrt{6}}{7}$ CAH $\cos(\alpha) = \frac{5}{7}$ TOA $\tan(\beta) = \frac{5}{2\sqrt{6}}$

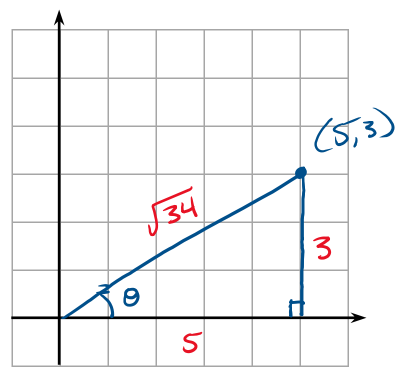
EXAMPLE 6

(a) Use the provided graph to plot the coordinate $(5,3)$. With this coordinate and the origin as vertices, construct a right triangle with base along the positive x-axis.

$c^2 = 5^2 + 3^2$

$c^2 = 34$

$c = \sqrt{34}$



(b) Let θ be the positive (*counterclockwise rotation*) acute angle measured from the positive x- axis (*initial side*) to the side formed by the origin and the given coordinate (*terminal side*). Evaluate each of the following.

SOH $\sin\theta = \frac{3}{\sqrt{34}}$ TOA $\tan\theta = \frac{3}{5}$

(c) Write an equation for the line passing through the origin and the given coordinate. Compare this equation with the values found in Part (b). What do you notice?

$y = mx + b$, $m = \text{slope} = \frac{\Delta y}{\Delta x} = \frac{\text{rise}}{\text{run}}$, $m = \frac{3}{5}$

$b = y\text{-value of } y\text{-int.}$, $b = 0$

$y = \frac{3}{5}x$

slope = tangent value!

EXAMPLE 7

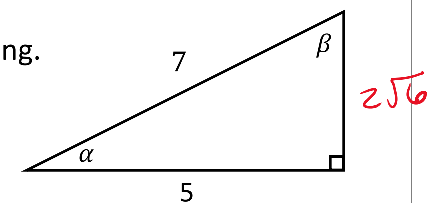
Using SOH-CAH-TOA, we can recall three of the six right triangle trigonometric ratios, namely sine, cosine, and tangent. Perhaps you've heard of three others, cosecant, secant, and cotangent, which are just reciprocals of the three we already know.

$$\text{sine: } \sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} \quad \rightarrow \quad \text{cosecant: } \csc(\theta) = \frac{\text{hyp}}{\text{opp}}$$

$$\text{cosine: } \cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \rightarrow \quad \text{secant: } \sec(\theta) = \frac{\text{hyp}}{\text{adj}}$$

$$\text{tangent: } \tan(\theta) = \frac{\text{oppostite}}{\text{adjacent}} \quad \rightarrow \quad \text{cotangent: } \cot(\theta) = \frac{\text{adj}}{\text{opp}}$$

(a) Using the provided triangle, evaluate each of the following.
(**Hint:** Look back at Example 5.)



$\sin(\alpha) = \frac{\text{opp}}{\text{hyp}} = \frac{2\sqrt{6}}{7}$

$\cos(\alpha) = \frac{\text{adj}}{\text{hyp}} = \frac{5}{7}$

$\csc(\alpha) = \frac{7}{2\sqrt{6}}$

$\sec(\alpha) = \frac{7}{5}$

$\cot(\beta) = \frac{2\sqrt{6}}{5}$

$\tan(\beta) = \frac{\text{opp}}{\text{adj}} = \frac{5}{2\sqrt{6}}$

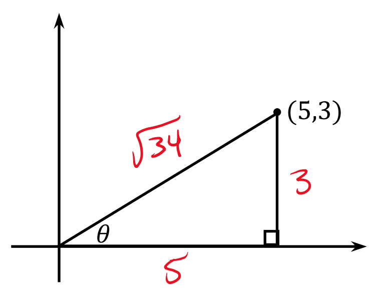
(b) Consider the right triangle in quadrant I shown here. Evaluate each of the following.
(**Hint:** Look back at Example 6.)

$\csc\theta = \frac{\sqrt{34}}{3}$

$\sin\theta = \frac{3}{\sqrt{34}}$

$\cot\theta = \frac{5}{3}$

$\tan\theta = \frac{3}{5}$



Looking Ahead…

In this lesson, we've seen several examples of an angle passing through a point in the xy-coordinate plane using the positive x-axis as a reference. This is an important concept that we'll fully develop later. For now, we note that any angle formed this way is said to be in ***standard position.***